

SPAM EMAIL CHECKER – PROJECT REPORT



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 - **Course: Fundamentals of AI & ML**
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1. Introduction

In today's digital era , email has become one of the most important ways of communication. From college updates to job opportunities, everything comes through email.

However, along with useful messages, we also receive a large number of spam emails  — such as fake offers, scam messages, and unwanted advertisements.

This project, Spam Email Checker, is a simple Machine Learning-based solution that helps in identifying whether an email is:

-  Legitimate (Not Spam / Ham)
-  Spam

The project focuses on understanding the core idea of text classification in a simple and practical way.

2. Motivation

The motivation behind this project comes from a very common real-life problem 🕯️

Almost everyone has experienced:

- **Receiving fake “You won a prize!” messages 📧**
- **Getting unwanted promotional emails 📧**
- **Missing important emails because of spam clutter 🕯️**

As a beginner in Machine Learning, I wanted to build something:

- **Simple yet meaningful**
- **Practical and relatable**
- **Useful in real-world scenarios**

Instead of directly jumping into complex AI systems, I decided to start with a basic spam detection system to understand:

- **How machines understand text 📧**
- **How models learn patterns 📈**
- **How predictions are made 📊**

This project is not just about coding — it represents my first step into solving real-world problems using AI 📧

🔄 3. Objective

The main objectives of this project are:

- **📧 Accept email text as input**

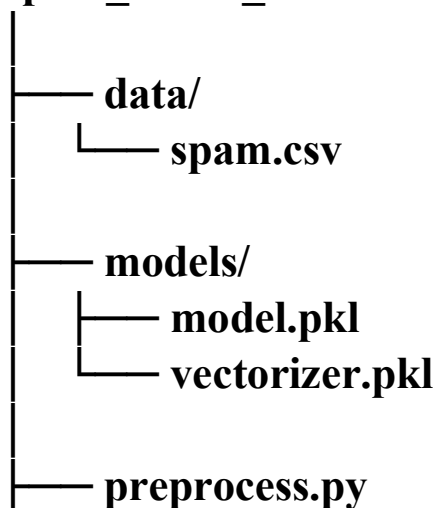
- 📄 Clean and preprocess the text
 - 📊 Convert text into numerical format
 - 🧠 Apply Machine Learning model
 - 📈 Predict and display result (Spam / Not Spam)
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🔧 4. Technologies Used

- 🐍 Python
 - 🏠 Scikit-learn
 - 🌀 Machine Learning (Naive Bayes)
 - 📦 TF-IDF Vectorizer
 - 🗣️ NLTK (Natural Language Toolkit)
 - 🥄 Pickle (for model storage)
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🗂️ 5. Project Structure

Spam_Email_Classifier/



├── train.py
├── email_checker.py
└── requirements.txt

⚙️ 6. Working of the Project

◆ Step 1: Data Collection

- The dataset contains labeled emails (spam / ham)

◆ Step 2: Text Preprocessing

- Convert text to lowercase
- Remove special characters
- Remove stopwords (common words like *the, is, at*)

◆ Step 3: Feature Extraction |

- TF-IDF is used to convert text into numbers
- Helps model understand importance of words

◆ Step 4: Model Training |

- Algorithm used: Naive Bayes
- Learns patterns from dataset

◆ Step 5: Prediction

- User enters email text
- Model predicts:
 - Spam 
 - Not Spam 

7. Implementation Overview

The project consists of three main parts:

Preprocessing Module

- Cleans and prepares the input text

Training Module

- Trains the ML model
- Saves it using pickle

Email Checker (CLI Tool)

- Takes user input
 - Predicts result instantly
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8. How to Run the Project

Step 1: Navigate to folder

`cd Spam_Email_Classifier`

Step 2: Install dependencies

`pip install -r requirements.txt`

Step 3: Train model

`python train.py`

Step 4: Run project

`python email_checker.py`

✈ 9. Sample Test Cases

+ Spam Example

Congratulations! You have won a free prize

■ Not Spam Example

Hi, are we meeting tomorrow for the project?

✂ 10. Results

- The model successfully identifies spam emails
 - Performs well on basic dataset
 - Gives quick predictions in real-time
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📊 11. Learning Outcomes

Through this project, I gained practical understanding of:

- Text preprocessing techniques 📄
- TF-IDF vectorization |
- Naive Bayes algorithm 📊
- Model training & saving 💻
- Building CLI-based applications 🖥

Most importantly, I learned how theory is applied in real-world projects.

■ 12. Limitations

- Dataset is limited 
 - Not highly accurate for complex emails
 - Works only as a basic model
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13. Future Scope

This project can be improved by:

-  Using Deep Learning (LSTM, BERT)
 -  Creating a web-based interface
 -  Adding accuracy and analytics dashboard
 -  Building a mobile application
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14. Conclusion

The Spam Email Checker project is a simple yet effective implementation of Machine Learning in solving a real-world problem.

This project helped me move from learning concepts → building something practical.

It marks an important step in my journey towards becoming a skilled developer in AI & ML 

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